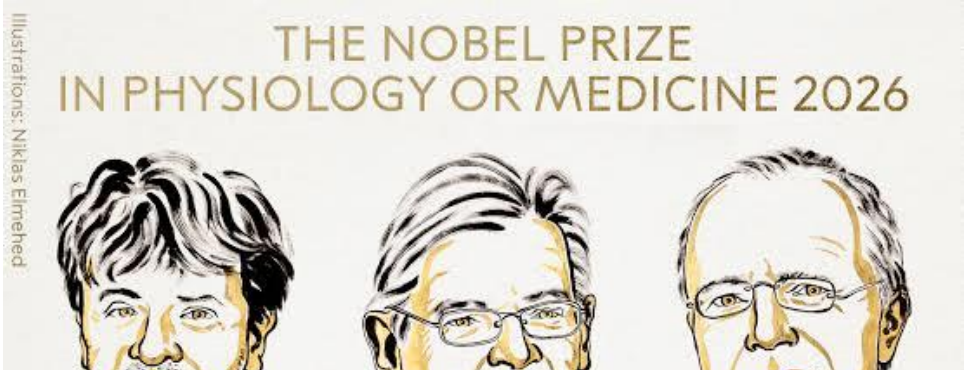
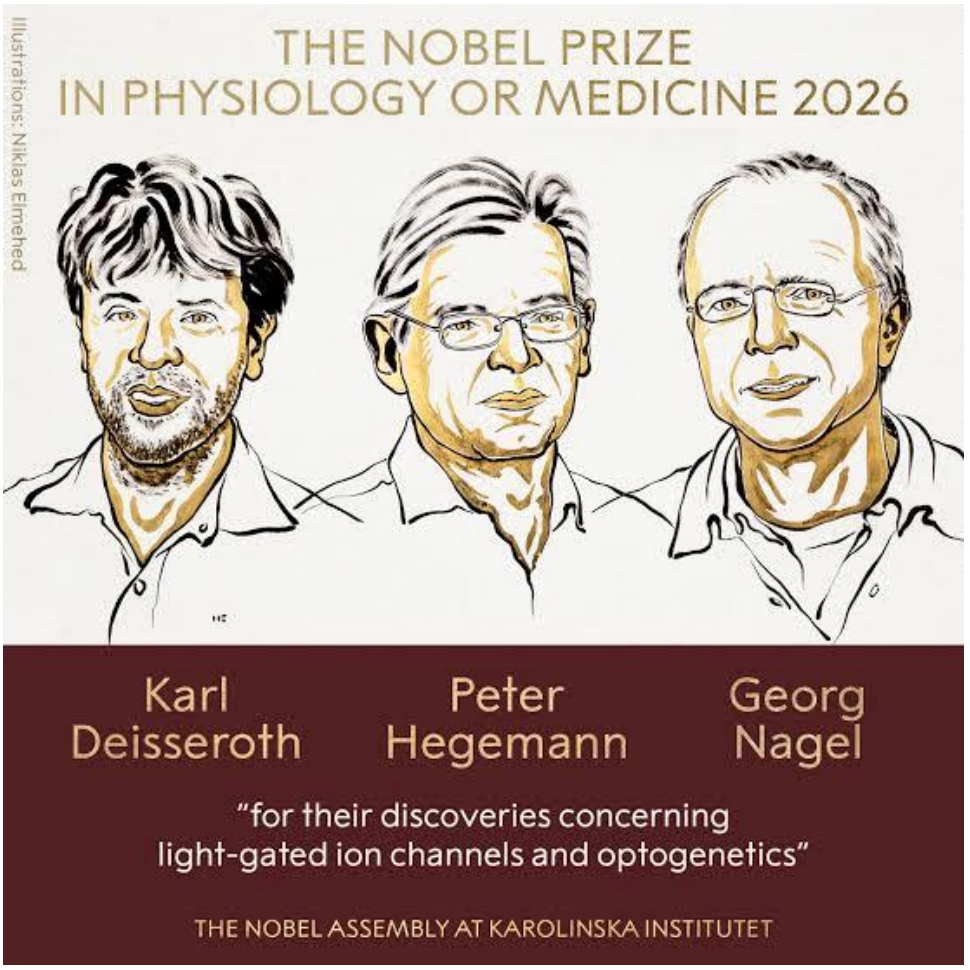
Application of Derivate: Max & Min

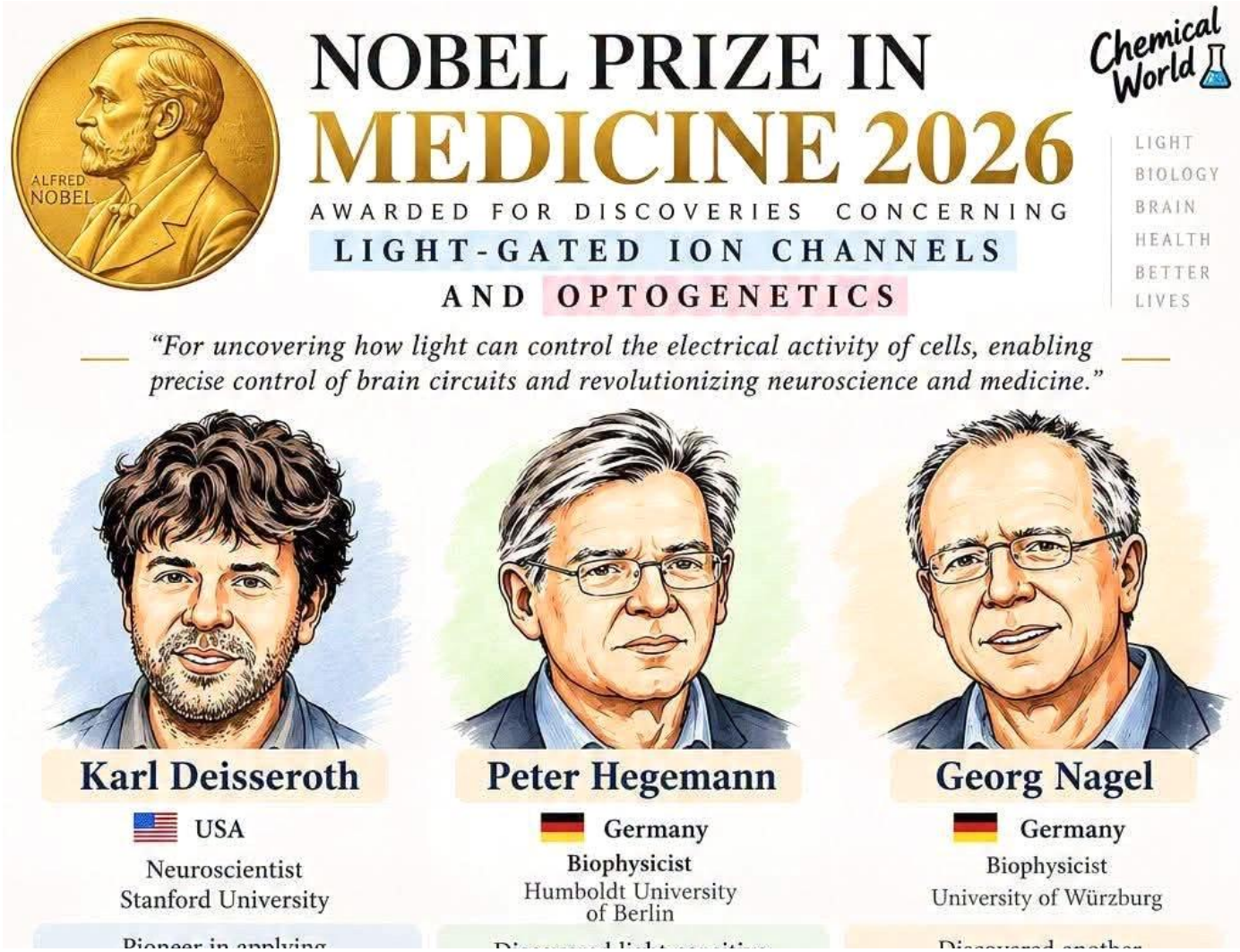
Monday, 28 September 2026 5:29 pm








Illustrations: Niklas Elmehed
THE NOBEL PRIZE
IN PHYSIOLOGY OR MEDICINE 2026
Karl Deisseroth
Peter Hegemann
Georg Nagel
"for their discoveries concerning light-gated ion channels and optogenetics"
THE NOBEL ASSEMBLY AT KAROLINSKA INSTITUTET

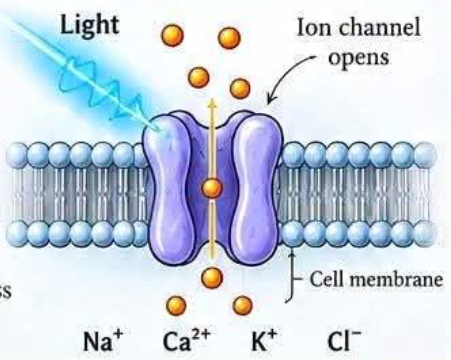

ALFRED NOBEL
NOBEL PRIZE IN
MEDICINE 2026
AWARDED FOR DISCOVERIES CONCERNING
LIGHT-GATED ION CHANNELS
AND OPTOGENETICS
Chemical World
LIGHT
BIOLOGY
BRAIN
HEALTH
BETTER
LIVES
"For uncovering how light can control the electrical activity of cells, enabling precise control of brain circuits and revolutionizing neuroscience and medicine."
Karl Deisseroth
USA
Neuroscientist
Stanford University
Peter Hegemann
Germany
Biophysicist
Humboldt University of Berlin
Georg Nagel
Germany
Biophysicist
University of Würzburg

Neuroscientist Stanford University	Biophysicist Humboldt University of Berlin	Biophysicist University of Würzburg
Pioneer in applying optogenetics to study and manipulate brain circuits.	Discovered light-sensitive ion channels in algae, including channelrhodopsin.	Discovered another light-gated ion channel, halorhodopsin.

LIGHT-GATED ION CHANNELS

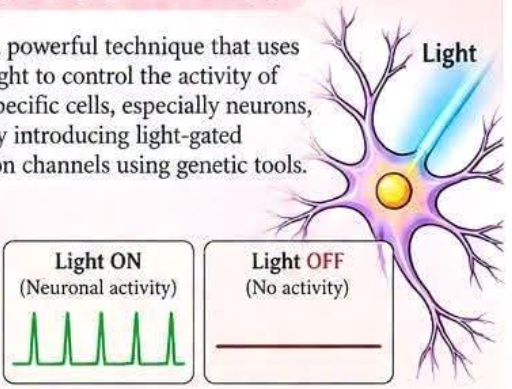
These are proteins in the cell membrane that open or close in response to light of a specific wavelength.

- Absorb light
- Change shape
- Open/close ion channel
- Allow specific ions to pass (Na^+, K^+, Ca^{2+}, Cl^-)



OPTOGENETICS

A powerful technique that uses light to control the activity of specific cells, especially neurons, by introducing light-gated ion channels using genetic tools.



MAJOR IMPACTS

Neuroscience
Understanding how brain circuits work.

Neurological Disorders
New approaches for epilepsy, Parkinson's, depression and more.

Vision Restoration
Potential therapies for blindness.

Drug Development
Better models to study brain function and diseases.

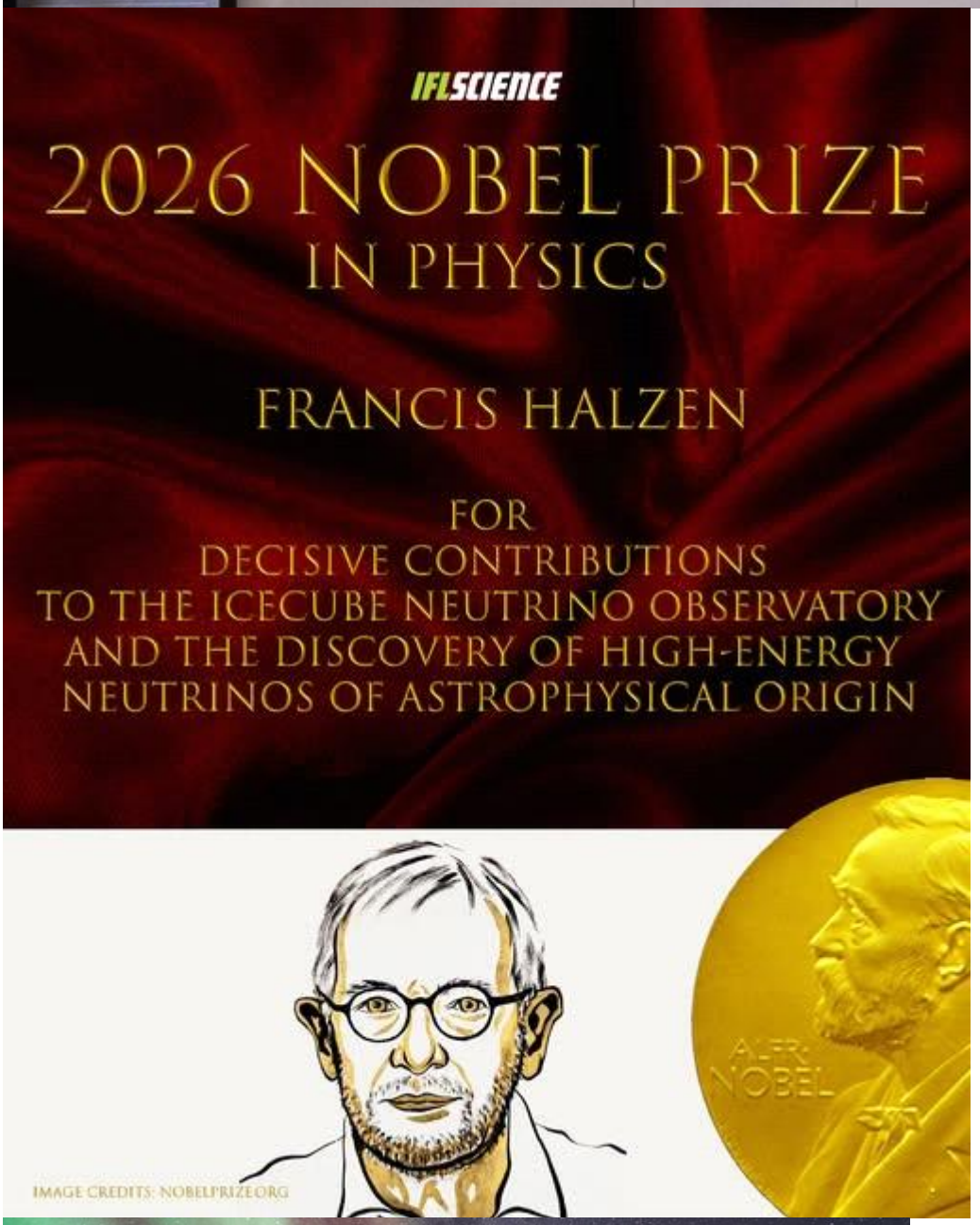
Mapping Neural Circuits
High-precision control of specific neurons in living brains.

Their pioneering work has transformed neuroscience and opened new possibilities for treating a wide range of diseases.

Chemical World

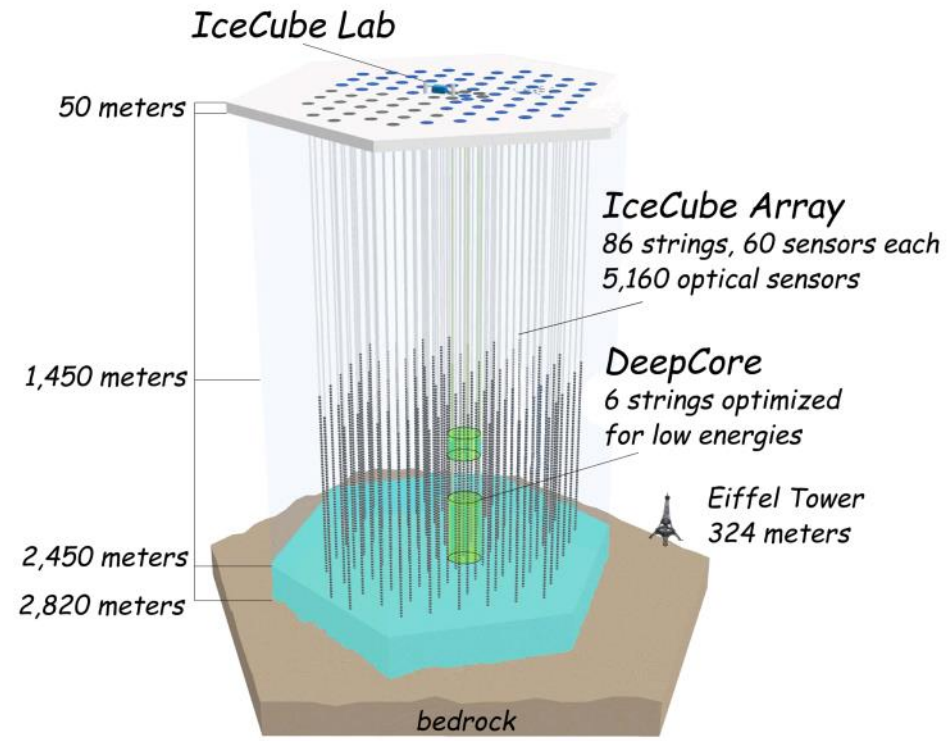




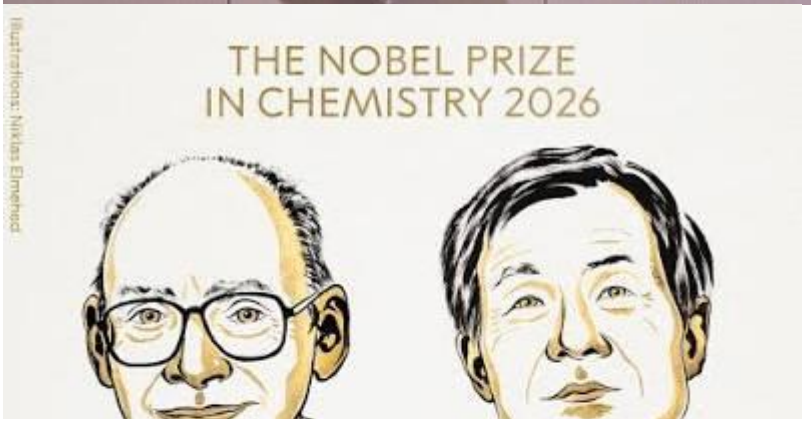

IFLSCIENCE
2026 NOBEL PRIZE
IN PHYSICS
FRANCIS HALZEN
FOR
DECISIVE CONTRIBUTIONS
TO THE ICECUBE NEUTRINO OBSERVATORY
AND THE DISCOVERY OF HIGH-ENERGY
NEUTRINOS OF ASTROPHYSICAL ORIGIN
IMAGE CREDITS: NOBELPRIZE.ORG
ALFRED NOBEL






IceCube Lab
50 meters
IceCube Array
86 strings, 60 sensors each
5,160 optical sensors
1,450 meters
DeepCore
6 strings optimized
for low energies
Eiffel Tower
324 meters
2,450 meters
2,820 meters
bedrock


NOBELPRISET I KEMI 2026
THE NOBEL PRIZE IN CHEMISTRY 2026
AKADEMIEN
"för upptäckt av icke-linjära effekter och autokatalys i asymmetrisk organisk syntes"
"for the discovery of non-linear effects and autocatalysis in asymmetric organic synthesis"
Henri Kagan
Université Paris-Sud, France
Born 1930, France
Kenso Soai
Tokyo University of Science, Japan
Born 1950, Japan


THE NOBEL PRIZE
IN CHEMISTRY 2026
Illustrations: Niklas Elmehed



Henri B. Kagan Kenso Soai

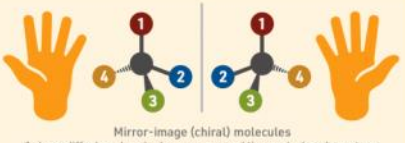
"for the discovery of non-linear effects and autocatalysis in asymmetric organic synthesis"

THE ROYAL SWEDISH ACADEMY OF SCIENCES

The 2026 Nobel Prize in Chemistry

The 2026 Nobel Prize in Chemistry was awarded to **Henri Kagan** and **Kenso Soai** for the discovery of non-linear effects and autocatalysis in asymmetric organic synthesis.

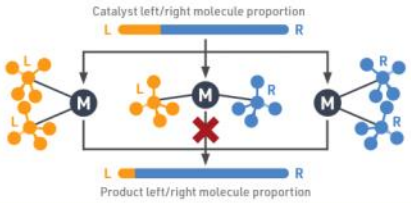
Many carbon-based molecules are chiral – they have mirror image forms, much like our hands. Almost all proteins in living organisms are built from left-handed versions of amino acids. However, lab reactions often produce both mirror images of molecules. This year's Nobel Prize goes to research on reactions that help chemists produce only one mirror image of a molecule.

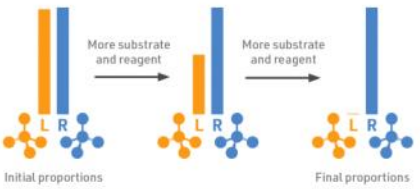


Mirror-image (chiral) molecules
1-4 are differing chemical groups around the central carbon atom.

In 1986, **Henri Kagan** found a way to boost the formation of one of the chiral products in chemical reactions. The catalyst for these reactions consists of a metal ion attached to a different chiral molecule. Kagan discovered that by attaching a mix of left and right molecules to the metal ion, he could product *non-linear effects* – that is, proportions of left and right product molecules differed from the proportion of the left and right molecules used with the catalyst.

In 1995, **Kenso Soai** designed the first ever lab-based chemical reaction which made only one of the product's possible mirror images. He reacted a non-mirror-image molecule to form a chiral product. Once formed, this product speeds up the creation of more of the same product – *autocatalysis*. By starting with a small excess of either the left or right molecules, this autocatalysis can eventually lead to the excess molecule forming 99.99% of the reaction product.



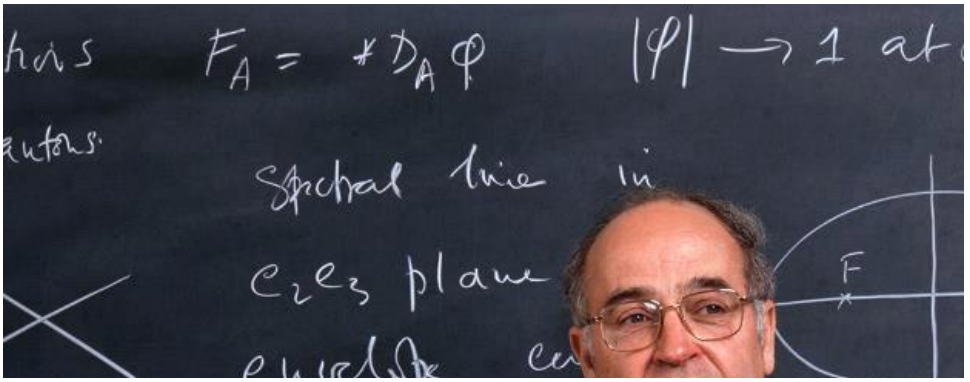


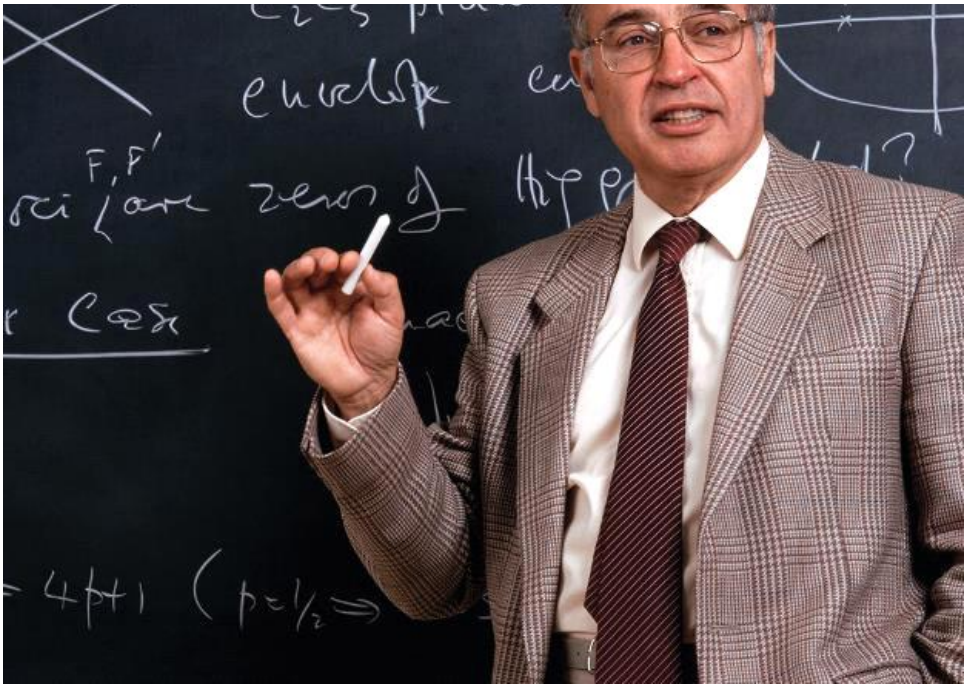
WHY DOES THIS RESEARCH MATTER?

The different mirror images of some molecules can have different properties. This can be of great importance for the synthesis of medicines, as well as providing insights into why biological molecules are formed from single mirror image building blocks.

Nobel Prize in Chemistry press release: https://www.nobelprize.org/prizes/chemistry/2026/press-release/

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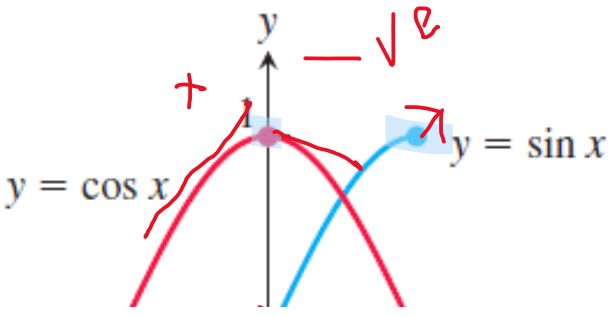


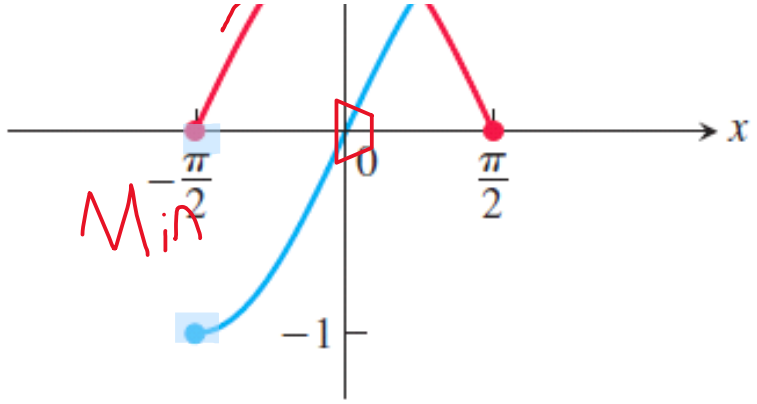


ALGEBRA IS THE OFFER MADE BY THE DEVIL TO THE MATHEMATICIAN. THE DEVIL SAYS: I WILL GIVE YOU THIS POWERFUL MACHINE, IT WILL ANSWER ANY QUESTION YOU LIKE. ALL YOU NEED TO DO IS GIVE ME YOUR SOUL: GIVE UP GEOMETRY AND YOU WILL HAVE THIS MARVELLOUS MACHINE'.

- MICHAEL ATIYAH -

LIBQUOTES.COM





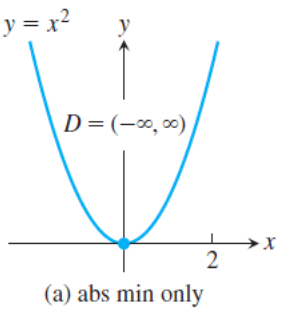
DEFINITIONS Let f be a function with domain D. Then f has an **absolute maximum** value on D at a point c if

$$f(x) \le f(c) \qquad \text{for all } x \text{ in } D$$

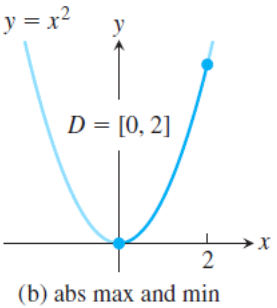
and an **absolute minimum** value on D at c if

$$f(x) \ge f(c) \qquad \text{for all } x \text{ in } D.$$

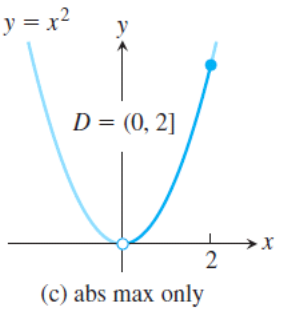
Function rule	Domain D	Absolute extrema on D
(a) $y = x^2$	$(-\infty, \infty)$	No absolute maximum Absolute minimum of 0 at $x = 0$
(b) $y = x^2$	$[0, 2]$	Absolute maximum of 4 at $x = 2$ Absolute minimum of 0 at $x = 0$
(c) $y = x^2$	$(0, 2]$	Absolute maximum of 4 at $x = 2$ No absolute minimum
(d) $y = x^2$	$(0, 2)$	No absolute extrema



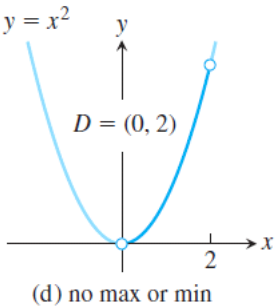
(a) abs min only



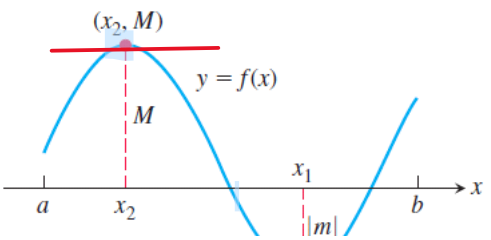
(b) abs max and min

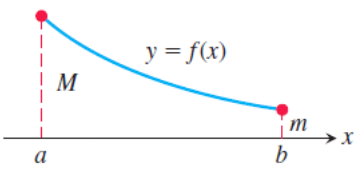


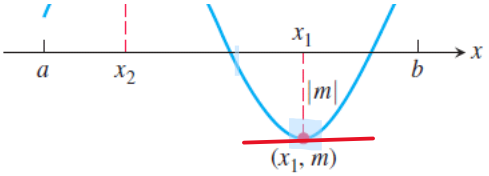
(c) abs max only



(d) no max or min



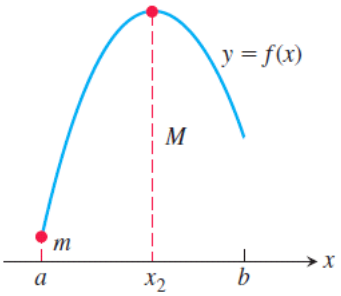




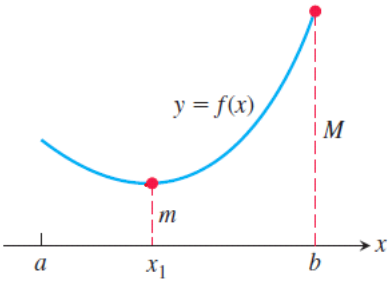
Maximum and minimum at interior points



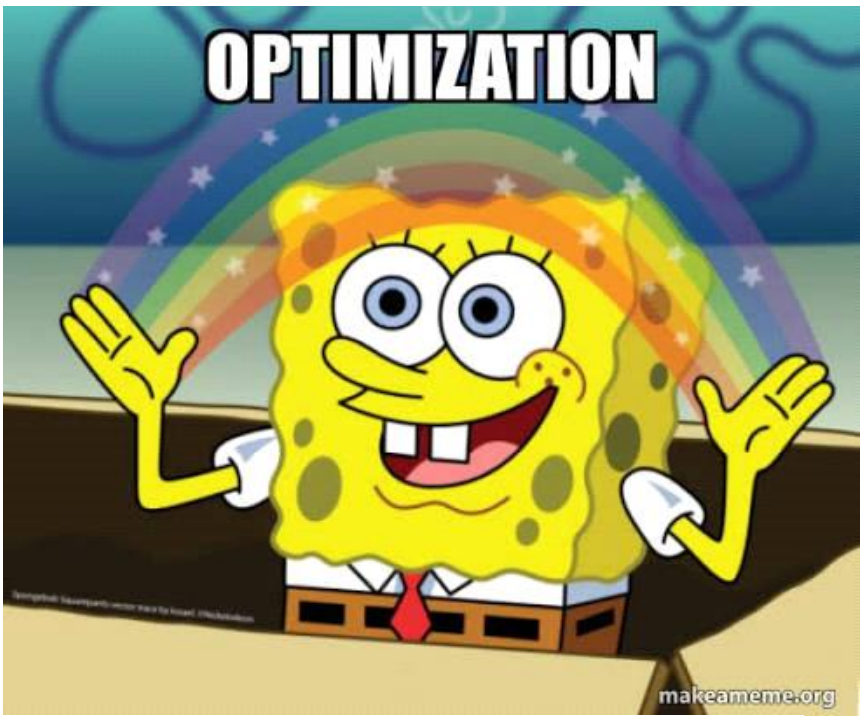
Maximum and minimum at endpoints



Maximum at interior point, minimum at endpoint



Minimum at interior point, maximum at endpoint

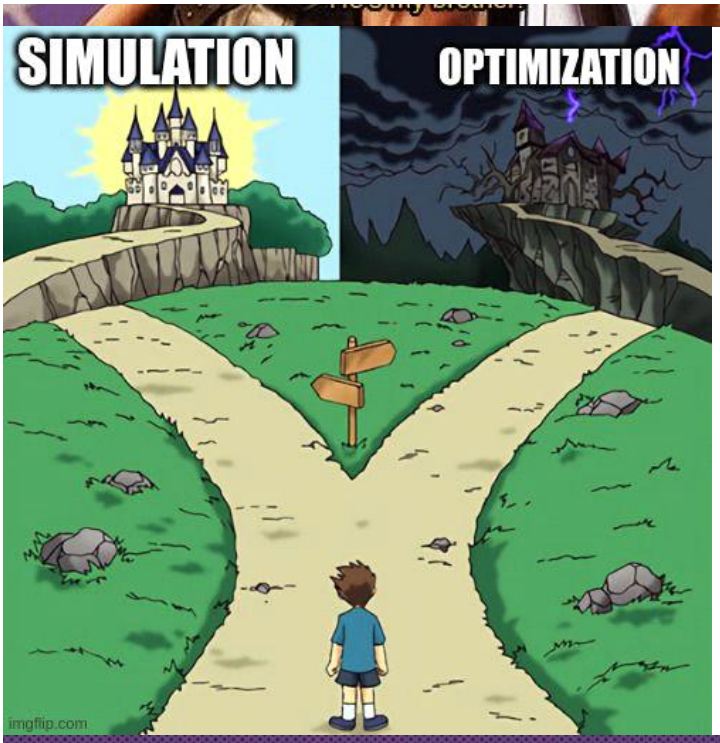





ALL THE THINGS
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OPTIMIZATION
OPTIMIZATION EVERYWHERE
makeameme.org

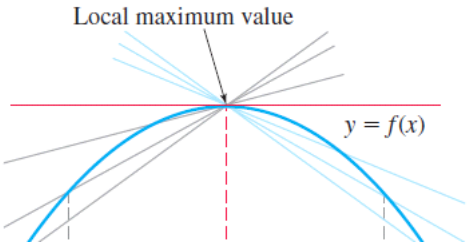

Every new game in these days:
Game developer
I've never met this man in my life.
Optimisation
He's my brother!
SIMULATION
OPTIMIZATION





THEOREM 2—The First Derivative Theorem for Local Extreme Values If f has a local maximum or minimum value at an interior point c of its domain, and if f' is defined at c, then

$$f'(c) = 0.$$



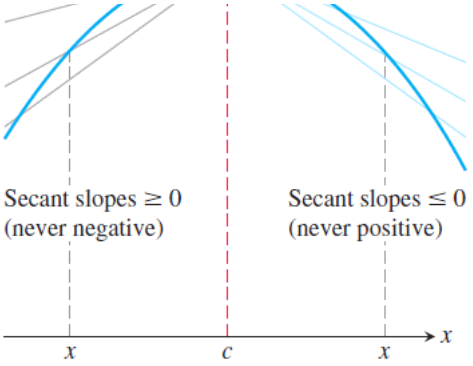
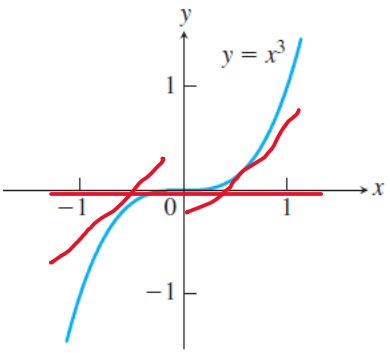


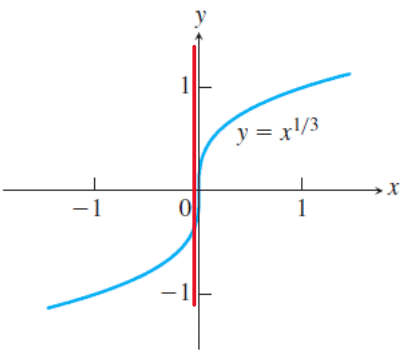
FIGURE 4.6 A curve with a local maximum value. The slope at c, simultaneously the limit of nonpositive numbers and nonnegative numbers, is zero.

DEFINITION An interior point of the domain of a function f where f' is zero or undefined is a **critical point** of f.

1. interior points where $f' = 0$,
2. interior points where f' is undefined,
3. endpoints of the domain of f.



(a)



(b)

FIGURE 4.7 Critical points without extreme values. (a) $y' = 3x^2$ is 0 at $x = 0$, but $y = x^3$ has no extremum there. (b) $y' = (1/3)x^{-2/3}$ is undefined at $x = 0$, but $y = x^{1/3}$ has no extremum there.

but $y = x^3$ has no extremum there.
(b) $y' = (1/3)x^{-2/3}$ is undefined at $x = 0$,
but $y = x^{1/3}$ has no extremum there.

DEFINITIONS A function f has a **local maximum** value at a point c within its domain D if $f(x) \le f(c)$ for all $x \in D$ lying in some open interval containing c.

A function f has a **local minimum** value at a point c within its domain D if $f(x) \ge f(c)$ for all $x \in D$ lying in some open interval containing c.

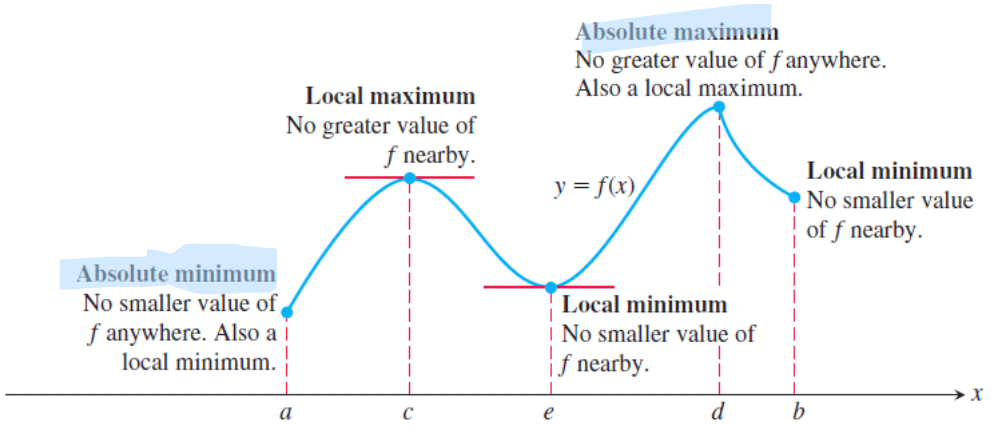


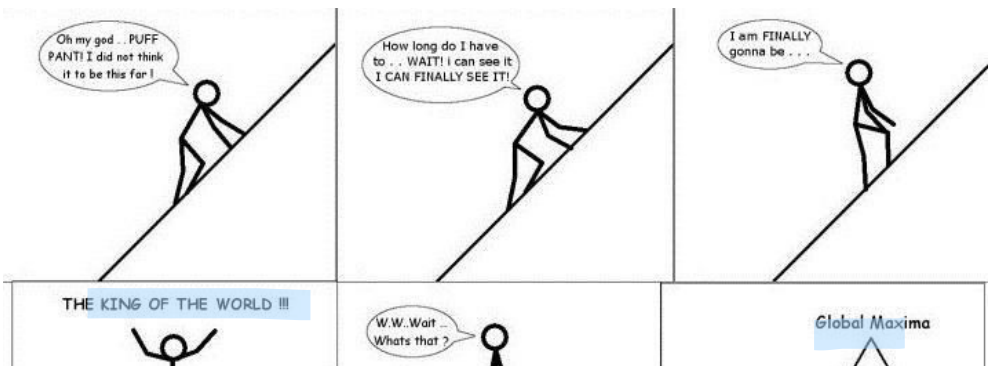
FIGURE 4.5 How to identify types of maxima and minima for a function with domain $a \le x \le b$.

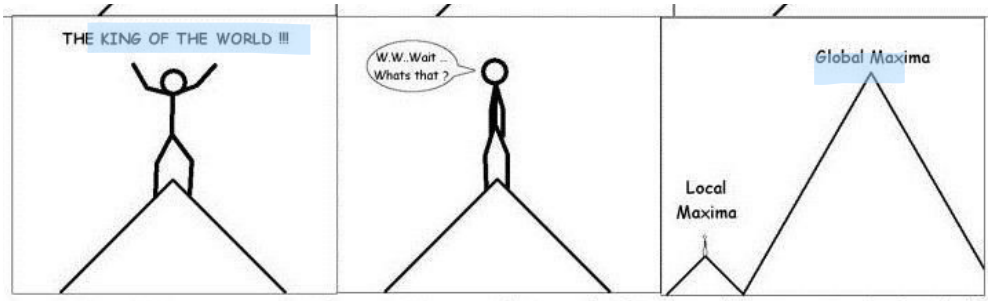
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nobody:

algorithm: ah yes the solution







Just because there is nothing left to derive in life $\frac{dl}{dt}=0$ doesnt mean you are the best . Know the difference. Dont be fooled

Me when I need to find the lowest possible point on the polynomial:




ABSOLUTE
MINIMA


GRADIENT DESCENT FINDS THE BEST SOLUTION
BE HONEST
IT FINDS A MINIMA
BE HONEST
IT FINDS A LOCAL MINIMA
THANK YOU


LEFT
EXIT 12
global minima
local minima
*gradient descent



How to Find the Absolute Extrema of a Continuous Function f on a Finite Closed Interval

1. Evaluate f at all critical points and endpoints.
2. Take the largest and smallest of these values.

EXAMPLE 2 Find the absolute maximum and minimum values of $f(x) = x^2$ on $[-2, 1]$.

Solution The function is differentiable over its entire domain, so the only critical point is where $f'(x) = 2x = 0$, namely $x = 0$. We need to check the function's values at $x = 0$ and at the endpoints $x = -2$ and $x = 1$:

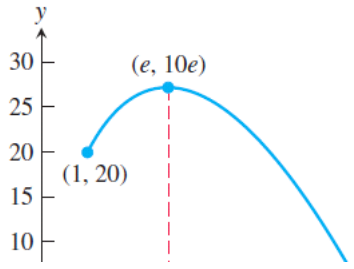
Critical point value: $f(0) = 0$

Endpoint values: $f(-2) = 4$

$f(1) = 1.$

The function has an absolute maximum value of 4 at $x = -2$ and an absolute minimum value of 0 at $x = 0$. ■

EXAMPLE 3 Find the absolute maximum and minimum values of $f(x) = 10x(2 - \ln x)$ on the interval $[1, e^2]$.



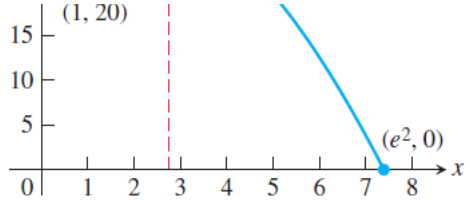


FIGURE 4.8 The extreme values of $f(x) = 10x(2 - \ln x)$ on $[1, e^2]$ occur at $x = e$ and $x = e^2$ (Example 3).

EXAMPLE 4 Find the absolute maximum and minimum values of $f(x) = x^{2/3}$ on the interval $[-2, 3]$.

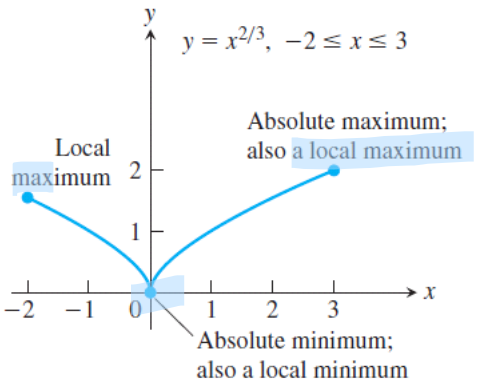
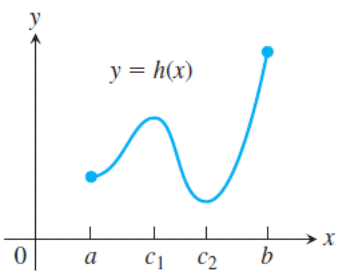


FIGURE 4.9 The extreme values of $f(x) = x^{2/3}$ on $[-2, 3]$ occur at $x = 0$ and $x = 3$ (Example 4).

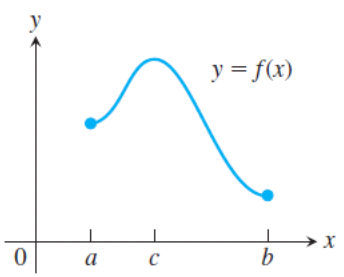
Finding Extrema from Graphs

In Exercises 1–6, determine from the graph whether the function has any absolute extreme values on $[a, b]$. Then explain how your answer is consistent with .Theorem 1

1.



2.



Absolute Extrema on Finite Closed Intervals

In Exercises 21–40, find the absolute maximum and minimum values of each function on the given interval. Then graph the function. Identify the points on the graph where the absolute extrema occur, and include their coordinates.

21. $f(x) = \frac{2}{3}x - 5, \quad -2 \le x \le 3$

24. $f(x) = 4 - x^3, \quad -2 \le x \le 1$

25. $f(x) = \frac{1}{\quad}, \quad 0.5 < x < 2$

24. $f(x) = 4 - x^3, \quad -2 \le x \le 1$

25. $F(x) = -\dfrac{1}{x^2}, \quad 0.5 \le x \le 2$

83. The function

$$V(x) = x(10 - 2x)(16 - 2x), \qquad 0 < x < 5,$$

models the volume of a box.

a. Find the extreme values of V.

b. Interpret any values found in part (a) in terms of the volume of the box.

Increasing Functions and Decreasing Functions

COROLLARY 3 Suppose that f is continuous on $[a, b]$ and differentiable on (a, b).

If $f'(x) > 0$ at each point $x \in (a, b)$, then f is increasing on $[a, b]$.

If $f'(x) < 0$ at each point $x \in (a, b)$, then f is decreasing on $[a, b]$.

EXAMPLE 1 Find the critical points of $f(x) = x^3 - 12x - 5$ and identify the open intervals on which f is increasing and on which f is decreasing.

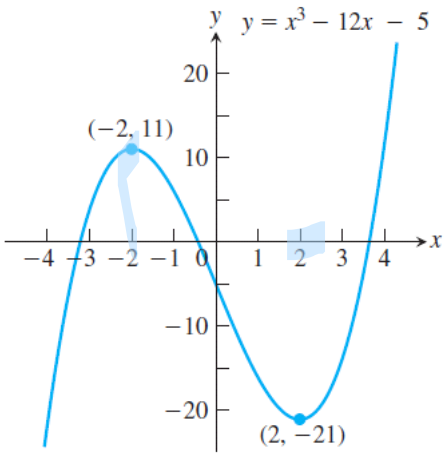


FIGURE 4.20 The function $f(x) = x^3 - 12x - 5$ is monotonic on three separate intervals (Example 1).

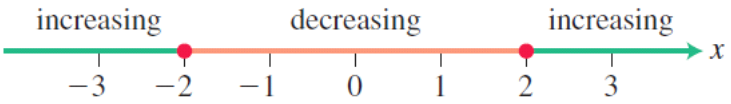
Solution The function f is everywhere continuous and differentiable. The first derivative

$$f'(x) = 3x^2 - 12 = 3(x^2 - 4)$$
$$= 3(x + 2)(x - 2)$$

is zero at $x = -2$ and $x = 2$. These critical points subdivide the domain of f to create non-overlapping open intervals $(-\infty, -2)$, $(-2, 2)$, and $(2, \infty)$ on which f' is either positive or negative. We determine the sign of f' by evaluating f' at a convenient point in each subin-

overlapping open intervals $(-\infty, -2)$, $(-2, 2)$, and $(2, \infty)$ on which f' is either positive or negative. We determine the sign of f' by evaluating f' at a convenient point in each subinterval. The behavior of f is determined by then applying Corollary 3 to each subinterval. The results are summarized in the following table, and the graph of f is given in Figure 4.20.

Interval	$-\infty < x < -2$	$-2 < x < 2$	$2 < x < \infty$
f' evaluated	$f'(-3) = 15$	$f'(0) = -12$	$f'(3) = 15$
Sign of f'	$+$	$-$	$+$
Behavior of f	increasing	decreasing	increasing



First Derivative Test for Local Extrema

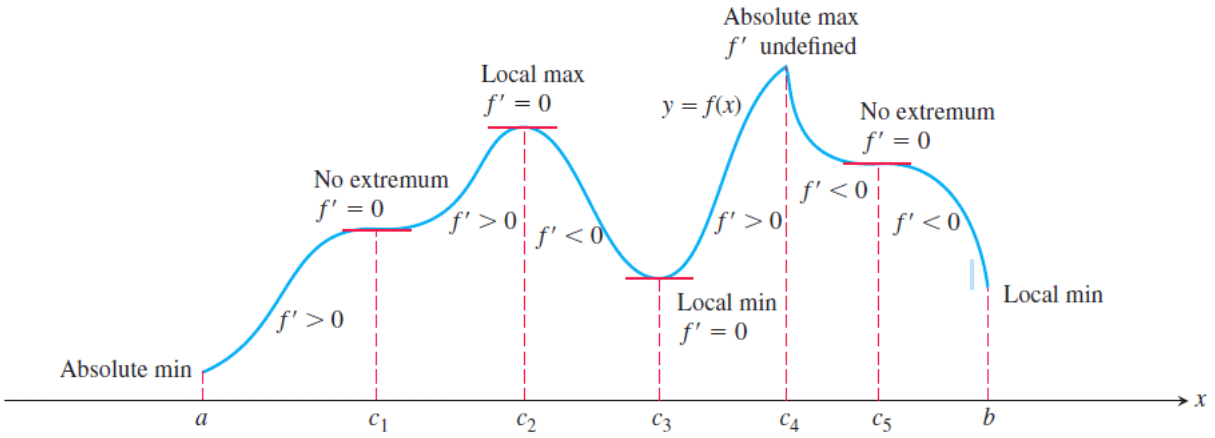


FIGURE 4.21 The critical points of a function locate where it is increasing and where it is decreasing. The first derivative changes sign at a critical point where a local extremum occurs.

First Derivative Test for Local Extrema

Suppose that c is a critical point of a continuous function f, and that f is differentiable at every point in some interval containing c except possibly at c itself. Moving across this interval from left to right,

1. if f' changes from negative to positive at c, then f has a local minimum at c;
2. if f' changes from positive to negative at c, then f has a local maximum at c;
3. if f' does not change sign at c (that is, f' is positive on both sides of c or negative on both sides), then f has no local extremum at c.

EXAMPLE 2 Find the critical points of

$$f(x) = x^{1/3}(x - 4) = x^{4/3} - 4x^{1/3}.$$

Identify the open intervals on which f is increasing and decreasing. Find the function's local and absolute extreme values.

Solution The function f is continuous at all x since it is the product of two continuous functions, $x^{1/3}$ and $(x - 4)$. The first derivative

$$f'(x) = \frac{d}{dx}(x^{4/3} - 4x^{1/3}) = \frac{4}{3}x^{1/3} - \frac{4}{3}x^{-2/3}$$

$$f'(x) = \frac{d}{dx}(x^{4/3} - 4x^{1/3}) = \frac{4}{3}x^{1/3} - \frac{4}{3}x^{-2/3}$$
$$= \frac{4}{3}x^{-2/3}(x - 1) = \frac{4(x - 1)}{3x^{2/3}}$$

Interval	$x < 0$	$0 < x < 1$	$x > 1$
Sign of f'	$-$	$-$	$+$
Behavior of f	decreasing	decreasing	increasing

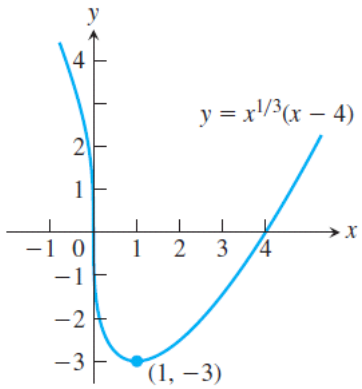


FIGURE 4.22 The function $f(x) = x^{1/3}(x - 4)$ decreases when $x < 1$ and increases when $x > 1$ (Example 2).

EXAMPLE 3 Find the critical points of

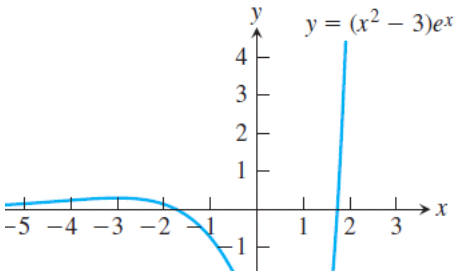
$$f(x) = (x^2 - 3)e^x.$$

$$f'(x) = (x^2 - 3) \cdot \frac{d}{dx}e^x + \frac{d}{dx}(x^2 - 3) \cdot e^x$$
$$= (x^2 - 3) \cdot e^x + (2x) \cdot e^x$$
$$= (x^2 + 2x - 3)e^x.$$

Since e^x is never zero, the first derivative is zero if and only if

$$x^2 + 2x - 3 = 0$$
$$(x + 3)(x - 1) = 0.$$

The zeros $x = -3$ and $x = 1$ partition the x-axis into open intervals as follows.



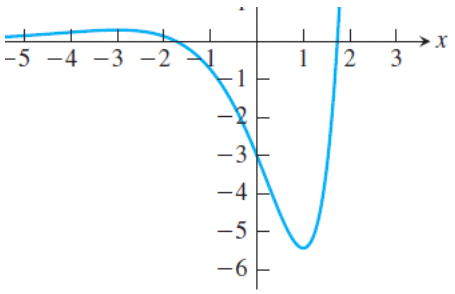
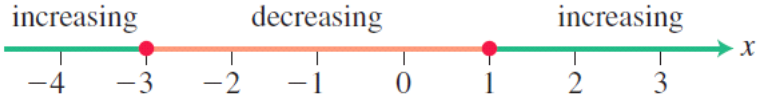


FIGURE 4.23 The graph of $f(x) = (x^2 - 3)e^x$ (Example 3).

Interval	$x < -3$	$-3 < x < 1$	$1 < x$
Sign of f'	+	−	+
Behavior of f	increasing	decreasing	increasing



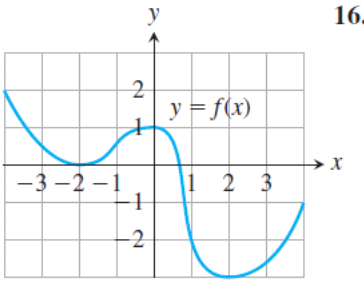
Identifying Extrema

In Exercises 15–44:

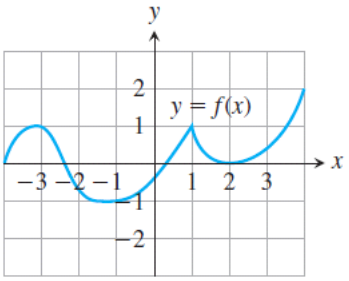
a. Find the open intervals on which the function is increasing and decreasing.

b. Identify the function's local and absolute extreme values, if any, saying where they occur.

15.



16.



Analyzing Functions from Derivatives

Answer the following questions about the functions whose derivatives are given in Exercises 1–14:

a. What are the critical points of f?

b. On what open intervals is f increasing or decreasing?

c. At what points, if any, does f assume local maximum and minimum values?

1. $f'(x) = x(x - 1)$

2. $f'(x) = (x - 1)(x + 2)$

3. $f'(x) = (x - 1)^2(x + 2)$

4. $f'(x) = (x - 1)^2(x + 2)^2$

5. $f'(x) = (x - 1)e^{-x}$

6. $f'(x) = (x - 7)(x + 1)(x + 5)$

7. $f'(x) = \dfrac{x^2(x - 1)}{x + 2}, \quad x \neq -2$

8. $f'(x) = \dfrac{(x - 2)(x + 4)}{(x + 1)(x - 3)}, \quad x \neq -1, 3$

9. $f'(x) = 1 - \dfrac{4}{x^2}, \quad x \neq 0$

10. $f'(x) = 3 - \dfrac{6}{\sqrt{x}}, \quad x \neq 0$

9. $f'(x) = 1 - \frac{}{x^2}, \quad x \neq 0$ **10.** $f'(x) = 3 - \frac{}{\sqrt{x}}, \quad x \neq 0$

Concavity and Curve Sketching

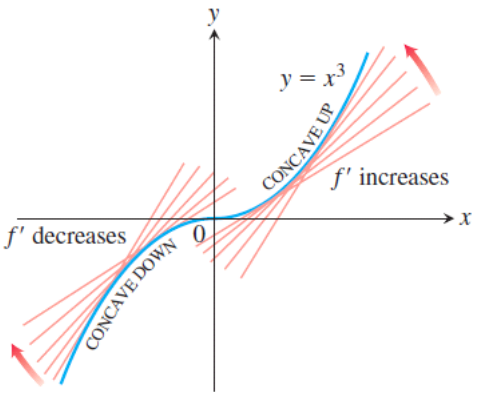


FIGURE 4.24 The graph of $f(x) = x^3$ is concave down on $(-\infty, 0)$ and concave up on $(0, \infty)$ (Example 1a).

DEFINITION The graph of a differentiable function $y = f(x)$ is

(a) concave up on an open interval I if f' is increasing on I;

(b) concave down on an open interval I if f' is decreasing on I.

The Second Derivative Test for Concavity

Let $y = f(x)$ be twice-differentiable on an interval I.

1. If $f'' > 0$ on I, the graph of f over I is concave up.

2. If $f'' < 0$ on I, the graph of f over I is concave down.

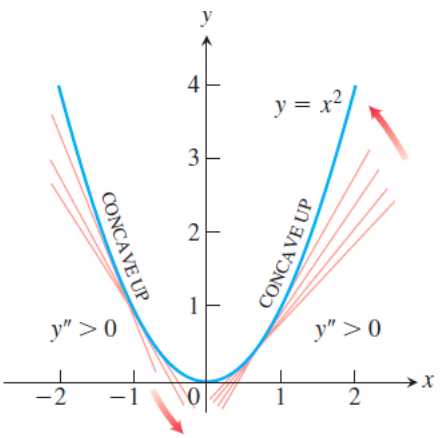


FIGURE 4.25 The graph of $f(x) = x^2$ is concave up on every interval (Example 1b).

EXAMPLE 1

(Example 10).

EXAMPLE 1

(a) The curve $y = x^3$ (Figure 4.24) is concave down on $(-\infty, 0)$ where $y'' = 6x < 0$ and concave up on $(0, \infty)$ where $y'' = 6x > 0$.

(b) The curve $y = x^2$ (Figure 4.25) is concave up on $(-\infty, \infty)$ because its second derivative $y'' = 2$ is always positive. ■

EXAMPLE 2

Determine the concavity of $y = 3 + \sin x$ on $[0, 2\pi]$.

Solution The first derivative of $y = 3 + \sin x$ is $y' = \cos x$, and the second derivative is $y'' = -\sin x$. The graph of $y = 3 + \sin x$ is concave down on $(0, \pi)$, where $y'' = -\sin x$ is negative. It is concave up on $(\pi, 2\pi)$, where $y'' = -\sin x$ is positive (Figure 4.26). ■

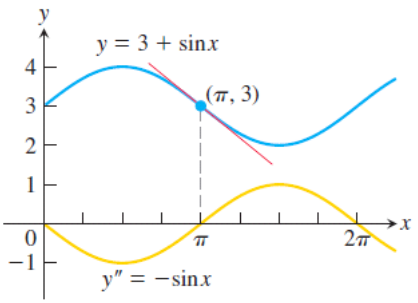


FIGURE 4.26 Using the sign of y'' to determine the concavity of y (Example 2).

Points of Inflection

The curve $y = 3 + \sin x$ in Example 2 changes concavity at the point $(\pi, 3)$. Since the first derivative $y' = \cos x$ exists for all x, we see that the curve has a tangent line of slope -1 at the point $(\pi, 3)$. This point is called a *point of inflection* of the curve. Notice from Figure 4.26 that the graph crosses its tangent line at this point and that the second derivative $y'' = -\sin x$ has value 0 when $x = \pi$. In general, we have the following definition.

DEFINITION A point $(c, f(c))$ where the graph of a function has a tangent line and where the concavity changes is a **point of inflection**.

At a point of inflection $(c, f(c))$, either $f''(c) = 0$ or $f''(c)$ fails to exist.

EXAMPLE 3

The graph of $f(x) = x^{5/3}$ has a horizontal tangent at the origin because $f'(x) = (5/3)x^{2/3} = 0$ when $x = 0$. However, the second derivative

$$f''(x) = \frac{d}{dx}\left(\frac{5}{3}x^{2/3}\right) = \frac{10}{9}x^{-1/3}$$

fails to exist at $x = 0$. Nevertheless, $f''(x) < 0$ for $x < 0$ and $f''(x) > 0$ for $x > 0$, so the second derivative changes sign at $x = 0$ and there is a point of inflection at the origin. The graph is shown in Figure 4.27. ■

y

The graph is shown in Figure 4.27.

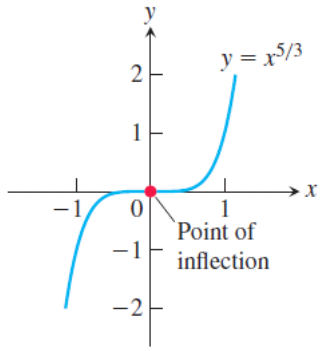


FIGURE 4.27 The graph of $f(x) = x^{5/3}$ has a horizontal tangent at the origin where the concavity changes, although f'' does not exist at $x = 0$ (Example 3).

EXAMPLE 4 The curve $y = x^4$ has no inflection point at $x = 0$ (Figure 4.28). Even though the second derivative $y'' = 12x^2$ is zero there, it does not change sign.

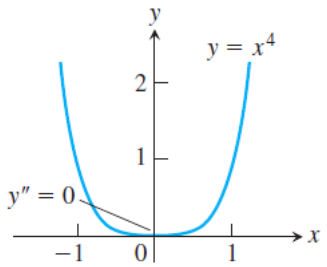


FIGURE 4.28 The graph of $y = x^4$ has no inflection point at the origin, even though $y'' = 0$ there (Example 4).

EXAMPLE 5 The graph of $y = x^{1/3}$ has a point of inflection at the origin because the second derivative is positive for $x < 0$ and negative for $x > 0$:

$$y'' = \frac{d^2}{dx^2}\left(x^{1/3}\right) = \frac{d}{dx}\left(\frac{1}{3}x^{-2/3}\right) = -\frac{2}{9}x^{-5/3}.$$

However, both $y' = x^{-2/3}/3$ and y'' fail to exist at $x = 0$, and there is a vertical tangent there. See Figure 4.29.

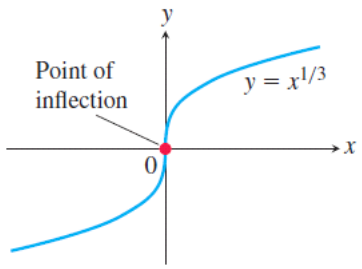


FIGURE 4.29 A point of inflection where y' and y'' fail to exist (Example 5)

EXAMPLE 6

where f' and f'' fail to exist (Example 5)

EXAMPLE 6 A particle is moving along a horizontal coordinate line (positive to the right) with position function

$$s(t) = 2t^3 - 14t^2 + 22t - 5, \qquad t \geq 0.$$

Find the velocity and acceleration, and describe the motion of the particle.

Solution The velocity is

$$v(t) = s'(t) = 6t^2 - 28t + 22 = 2(t - 1)(3t - 11),$$

and the acceleration is

$$a(t) = v'(t) = s''(t) = 12t - 28 = 4(3t - 7).$$

Interval	$0 < t < 1$	$1 < t < 11/3$	$11/3 < t$
Sign of $v = s'$	+	−	+
Behavior of s	increasing	decreasing	increasing
Particle motion	right	left	right

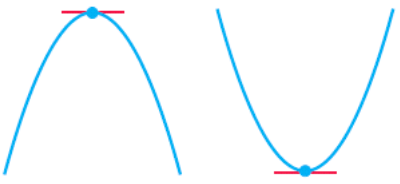
The particle is moving to the right in the time intervals $[0, 1)$ and $(11/3, \infty)$, and moving to the left in $(1, 11/3)$. It is momentarily stationary (at rest) at $t = 1$ and $t = 11/3$.

The acceleration $a(t) = s''(t) = 4(3t - 7)$ is zero when $t = 7/3$.

Interval	$0 < t < 7/3$	$7/3 < t$
Sign of $a = s''$	−	+
Graph of s	concave down	concave up

THEOREM 5—Second Derivative Test for Local Extrema Suppose f'' is continuous on an open interval that contains $x = c$.

1. If $f'(c) = 0$ and $f''(c) < 0$, then f has a local maximum at $x = c$.
2. If $f'(c) = 0$ and $f''(c) > 0$, then f has a local minimum at $x = c$.
3. If $f'(c) = 0$ and $f''(c) = 0$, then the test fails. The function f may have a local maximum, a local minimum, or neither.



$f' = 0, f'' < 0$ $\Rightarrow$ local max

$f' = 0, f'' > 0$ $\Rightarrow$ local min

EXAMPLE 7 Sketch a graph of the function

$$f(x) = x^4 - 4x^3 + 10$$

EXAMPLE 7 Sketch a graph of the function

$$f(x) = x^4 - 4x^3 + 10$$

using the following steps.

(a) Identify where the extrema of f occur.

(b) Find the intervals on which f is increasing and the intervals on which f is decreasing.

(c) Find where the graph of f is concave up and where it is concave down.

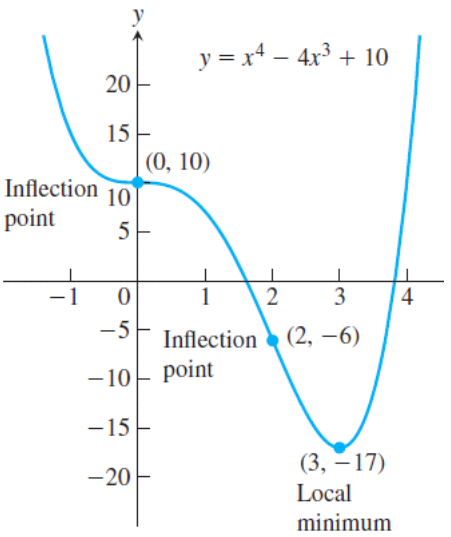
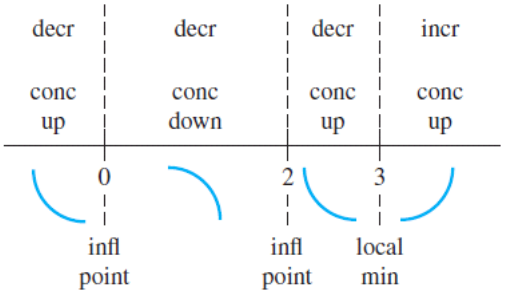
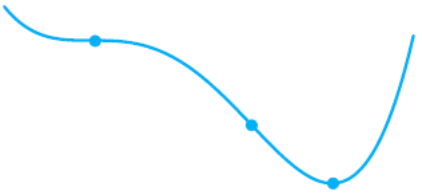


FIGURE 4.30 The graph of $f(x) = x^4 - 4x^3 + 10$ (Example 7).

The general shape of the curve is shown in the accompanying figure.



General shape



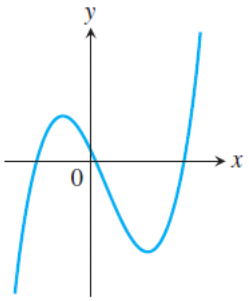
Procedure for Graphing $y = f(x)$

1. Identify the domain of f and any symmetries the curve may have.
2. Find the derivatives y' and y''.
3. Find the critical points of f, if any, and identify the function's behavior at each one.
4. Find where the curve is increasing and where it is decreasing.
5. Find the points of inflection, if any occur, and determine the concavity of the curve.
6. Identify any asymptotes that may exist.
7. Plot key points, such as the intercepts and the points found in Steps 3–5, and sketch the curve together with any asymptotes that exist.

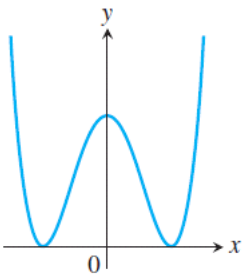
Analyzing Functions from Graphs

Identify the inflection points and local maxima and minima of the functions graphed in Exercises 1–8. Identify the intervals on which the functions are concave up and concave down.

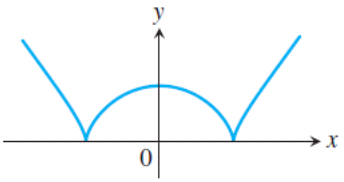
1. $y = \frac{x^3}{3} - \frac{x^2}{2} - 2x + \frac{1}{3}$



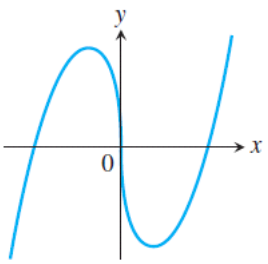
2. $y = \frac{x^4}{4} - 2x^2 + 4$



3. $y = \frac{3}{4}(x^2 - 1)^{2/3}$



4. $y = \frac{9}{14}x^{1/3}(x^2 - 7)$



In Exercises 9–58, identify the coordinates of any local and absolute extreme points and inflection points. Graph the function.

9. $y = x^2 - 4x + 3$

10. $y = 6 - 2x - x^2$

11. $y = x^3 - 3x + 3$

12. $y = x(6 - 2x)^2$